

An Improved Genetic Algorithm for Multiple Traveling Salesman Problem

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Abstract—Multiple traveling salesman problem, which uses the shortest total route as an optimization criteria, has huge application in both theoretical research and industry. This paper presents an improved genetic algorithm to provide an alternative and effective solution to the problem. The initial population was generated by greedy strategy, this enabled selected sub-route to be included in the initial population. Convergent speed was increased and at the same time complexity was significantly reduced. The mutation operator combined with 2-opt local search algorithm was used to avoid the limitation of local search ability of genetic algorithm. It also solved the problems of the simple genetic algorithm such as premature phenomena and slow convergence. The simulation results based on our algorithm show that the improved method is effective and feasible.

Index Terms—genetic algorithm, Multiple Traveling Salesman Problem, 2-opt local search algorithm

I. INTRODUCTION

Traveling Salesman Problem (TSP) is a problem that has widespread practical background and important theory value. With the increase of limiting factors, Multiple Traveling Salesman Problem (MTSP) can be derived from TSP. MTSP is the extension and continuation of the classic TSP, it discusses the optimization of the route arrangement for number M traveling salesmen to visit number N cities, the criteria is to arrange all traveling salesmen to start traveling from the same city (or different ones), they must use different traveling routes, so that each city in question is only visited by one traveling salesman (except the starting city), all traveling salesmen have to come back to the starting city at the end of their traveling and fulfil the criteria that the total routes are the shortest routes. MTSP is particularly important in practice, it is widely used in the fields of vehicle scheduling, pipelining, robot operation research, production scheduling and design of computer network topology.

In recent decades, research on MTSP has been developed to generate many effective and creative ways to solve the problem. For example, Prof. Holland at Michigan University introduced a genetic algorithm (GA), it is a highly parallel, random and self-adaptive search algorithm based on the mechanism of the natural selection and genetics [1]. This method is proven effective and widely used in TSP and MTSP due to its robustness.

Strong global searching ability is the advantage of simple genetic algorithm, but it has some limitations such as slow

convergence and falling into local minimum. In this paper we developed a method based on genetic algorithm. The initial population was generated by greedy strategy. Partially matched crossover (PMX) was selected as crossover operator. The mutation operator combined with 2-opt local searching algorithm was used to avoid the limitation of local search ability of genetic algorithm. It can to a certain degree solve the problems of the simple genetic algorithm such as premature phenomena and slow convergence.

II. DESCRIPTION OF MTSP

This paper focuses on the MTSP from the same starting city. MTSP is simplified to TSP by adding virtual cities, i.e. by introducing $M - 1$ virtual cities, named as $N+1, N+2, \dots, N+M - 1$. The traveling salesman is going back to the starting city if he meets a virtual city on his visiting route, so that to form a loop. Each loop represents a visiting route. The distance between the virtual cities and to starting city is defined to be infinity, so that total visiting routes M must include at least one city other than the starting city.

Based on the fact that each traveling salesman must visit at least one city, total traveling salesmen M have a total visiting route number of $N+M - 1$, the total distance of the traveling of MTSP is larger than that of TSP. In order to shorten the total traveling distance, total numbers of traveling salesmen $M - 1$ have to choose $M - 1$ cities closest to the starting city as their traveling routes, visiting routes to the rest $N - (M - 1)$ cities can then be solved by TSP.

III. DESIGN OF THE IMPROVED GENETIC ALGORITHM

A. Chromosome coding for MTSP

There are various coding methods for solving TSP, such as order expression, close neighbor expression, side expression and matrix expression, etc.[2] This paper uses order expression as the genetic coding, it means that genes are arranged by visiting cities, represented by numbers 0 to $N+M - 1$ and form a chromosome as follows:

$$[i_0, i_1, \dots, i_{N+M-1}]$$

Each number is only used once, meaning a traveling salesman is visiting city i_0 to i_1 until i_{N+M-1} . This method is simple, straightforward and easy to be understood.

B. Population initialization

Genetic algorithm is an operation on evolution of the initial population. Population initialization is the starting point of the calculation. In order to achieve the global optimum, initial population should be spread randomly in each area within the search space. Although random creation can have a diversity of the routes, it also has a low initial adaptability; this may affect directly the convergence speed. We in this paper use greedy strategy [3] to include as many optimized sub-routes as possible in the initial population. This method increased the adaptability in the initial population and reduced the blindness of the creation of the initial population.

Method of creation of the initial population is as follows.

Step 1: Select randomly a city labeled as A, if it is not selected before, mark it as selected, otherwise repeat the selection, then choose city B which is the closest city to A. City A and B can form a route AB.

Step 2: Choose separately Cities C and D which are the closest cities to A and B. C and D have to be cities not selected before. If $AC < BD$, then choose C to form route C-A-B, Otherwise, choose D to form route A-B-D.

Step 3: Repeat step 2 until all cities are included in traveling routes.

Step 4: If all cities have been selected at least once, choose randomly two cities to form a route and mark them as double selection, then go to step 2 until reach the requirement of the population.

C. Design of fitness function

Fitness function is used in genetic algorithm to judge each route, so that to decide the chance it can be passed to next generation, it is the limiting factor to the efficiency of evolution. This paper uses the distance of the route as the evaluation criteria, the fitness function is defined as the inverse of the total distance of the route, i.e. the shorter the route, the better the fitness.

D. The genetic operators for MTSP

1) *Selection operator*: Elitist strategy [4] proposed by De Jong K. A. was adopted for selection operation. This strategy copies the individual who has the best fitness at present to the next generation and, at the same time, removes the worst one in the next generation. The elitist strategy can keep the optimized individual at present in genetic algorithm and ensure that it is not lost or destroyed by selection, crossover and mutation. It is a critical ensurance to the convergence of genetic algorithm. Genetic algorithm that includes elitist strategy is globally convergent [5].

2) *Crossover operator*: Crossover operation exchanges part of the genes from two parent individuals to form two new individuals. It is an important character that distinguish genetic algorithm from other algorithms. In MTSP crossover operator can use partially matched crossover (PMX), ordered crossover (OX), cycle crossover (CX), etc.

This paper uses PMX operator introduced for TSP by Goldberg et al. in 1985. It requires random selection of two

crossing points to define a matching section. Based on the relationship from the middle parts between two crossing points in two parents, two new offspring were produced. For example, if there are two parents:

P1: 8 6 | 7 5 4 9 | 2 1 3

P2: 3 9 | 5 4 8 1 | 6 2 7

Firstly, we randomly select two crossing points, then exchange the middle part between two crossing points to make

O1 : X X | 5 4 8 1 | X X X

O2 : X X | 7 5 4 9 | X X X

Mapping relationship of the middle part is

$7 \longleftrightarrow 5$, $5 \longleftrightarrow 4$, $4 \longleftrightarrow 8$, $9 \longleftrightarrow 1$, X represents undetermined code.

For example, if using O2 to determine X, we firstly keep the non-visited cities numbered 3, 6, 2 inherited from its parents, this obtained

O2 : 3 X | 7 5 4 9 | 6 2 X

Then we use the original code of 9 in the parent P2 for the first X, and exchange $9 \longleftrightarrow 1$ to determine that it is 1. Using the same way to determine the second X, the original code is 7, exchange $7 \longleftrightarrow 5$ to determine that it is 5, but there is a 5 in O2 already, we exchange $5 \longleftrightarrow 4$ to determine that is 4, The same reason that O2 has 4 already, we exchange $4 \longleftrightarrow 8$ to determine it as 8, and the final results is

O2 : 3 1 7 5 4 9 6 2 8

3) *Mutation operator*: Mutation operation plays an import role in improving local search ability and maintaining variability of the population. It also prevents the premature in genetic algorithm. This paper uses 2-opt local search algorithm [6] to optimize 20% of the chromosomes in the population, it can both exchange two sides for mutation and be used as local search algorithm. This method has to some extent solved the problem of local search ability in genetic algorithm. At the same time, in order to keep the global search ability in genetic algorithm, the other chromosomes adopt inverse mutation .i.e. randomly select two different points, inverse the order of route formed by the gene between two points, this can form the new chromosome after mutation.

The principle of 2-opt local search algorithm is shown in Figure 1. It optimizes routes through each side and reverses sub-route. Circles without numbers represent a series routes between two or more cities. If $L_{12} + L_{34} > L_{13} + L_{24}$, L_{12} and L_{34} is replaced by L_{13} and L_{24} and the routes between city 2 and 3 are reversed in order. In general, 2-opt algorithm starts from a certain point and is followed by operation to two sides as mentioned above.

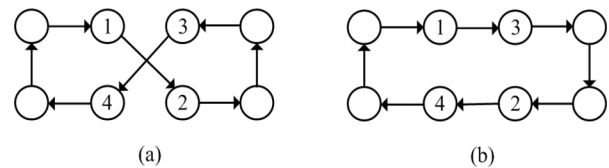


Figure 1. Schematic diagram of 2-opt local search algorithm

E. Termination conditions of the algorithm

This paper uses a combined rule of fixed evolution generations and pre-set convergent conditions to terminate the algorithm. If the pre-set convergent condition is fulfilled, the algorithm terminates. If it reaches the fixed evolution generations but still does not fulfill the pre-set convergent conditions, algorithm terminates as well. The pre-set convergent conditions are as follows, if the evolution reaches five generations, but the optimal solution is still not changed and the improvement of the fitness function is not better than 1%.

IV. SIMULATION RESULTS

Parameters chosen in genetic algorithm are very critical. They have direct impact on the convergence and quality of the final results. Due to the optimization in the initial population in this paper, the population can take a relatively low value, $POPNUM=80$. The termination of generation can be $G=5000$. Due to a series of the design of crossover and mutation in genetic algorithm, this ensures effectiveness of individual route, crossover probability p_c and mutation probability p_m can be set to relatively high values ($p_c = 0.9$ and $p_m = 0.1$) to increase the global search ability.

To show the correctness and effectiveness of our improved genetic algorithm, this paper simulates 5 traveling salesmen and numbers of cities are chosen to be 30, 50 and 100. Simulation results are shown in Figure 2, 3 and 4. (a) and (b) represent the results and the convergence curve of the improved genetic algorithm respectively.

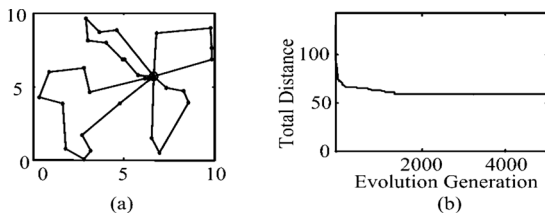


Figure 2. Simulation results of 30 cities

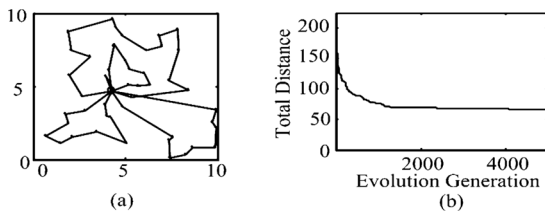


Figure 3. Simulation results of 50 cities

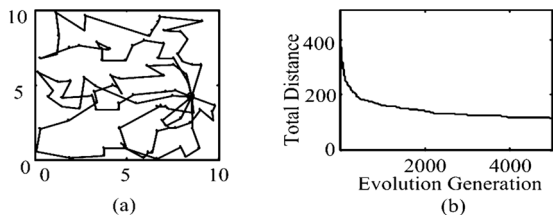


Figure 4. Simulation results of 100 cities

Figure 2, 3 and 4 show that it only takes 1148, 1629 and 2873 generations to reach the best MTSP results, and the corresponding shortest route are 58.9389, 67.3776 and 115.5245.

V. CONCLUSIONS

This paper provides an improved genetic algorithm to solve the MTSP. The initial population was generated by greedy strategy, it enabled selected sub-route to be included in the initial population. Convergent speed was increased and at the same time complexity was significantly reduced. The mutation operator combined with 2-opt local search algorithm was used to avoid the limitation of local search ability of genetic algorithm. To a certain extent, it solved the problems of the simple genetic algorithm such as premature phenomena and slow convergence. The simulation results based on our algorithm show that the improved method is effective and feasible.

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